ASSIGNMENT-2

1. Find the matrix representation of the Linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^2$ defined as $T(x, y, z) = (2x + 3y - z, 4x - y + 2z)$ with respect to the bases $B = \{(1, 1, 0), (1, 2, 3), (1, 3, 5)\}$ and $B' = \{(1, 2), (2, 3)\}$.

2. Consider the bases $S = \{(1,2), (2,3)\}$ and $S' = \{(1,3), (1,4)\}$ of $\mathbb{R}^2$. Find the change of basis matrix: (a) P from S to S' (b)Q from S' to S.

3. Find the angle θ between $u = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$ and $v = \begin{bmatrix} 0 & -1 \\ 2 & 3 \end{bmatrix}$.